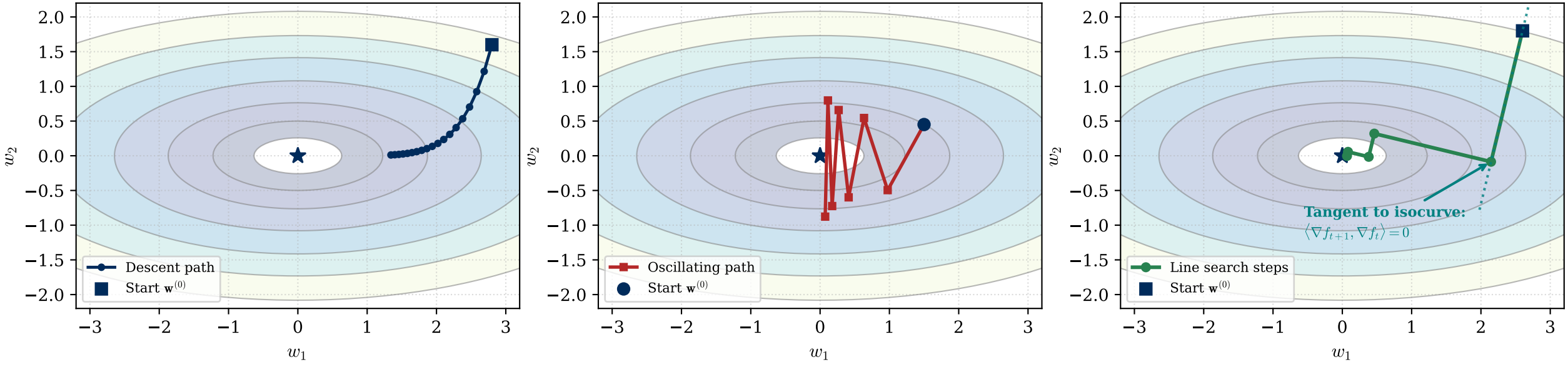

(a) Under-stepping ($\alpha \ll \alpha_{\text{opt}}$)
Slow crawling progress
(b) Overshooting ($\alpha > 2/\lambda_{\max}$)
Oscillates and diverges
(c) Exact Line Search ($\alpha = \alpha^*$)
Step lands tangent to isocurve
Descent path
Start $\mathbf{w}^{(0)}$
Oscillating path
Start $\mathbf{w}^{(0)}$
Line search steps
Start $\mathbf{w}^{(0)}$
Tangent to isocurve:
$\langle \nabla f_{t+1}, \nabla f_t \rangle = 0$
w_1
w_2